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Climate commoning and infrastructural care

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Abstract: Contemporary local urban governments struggle to approach ambitious environmental goals in the global climate crisis. As the understanding of the necessity to increase just and fair citizen engagement in governmental sustainability initiatives has become fundamental, the challenge emerges on how processes can become resilient beyond specific project initiatives. The concept of *infrastructruring*, which here is understood as a situated effort to hold together socio-material constellations that enable certain ways of acting in the world, provides a foundation for long-term interaction between the public sector and citizens. But as the participatory design community is strong on providing ”points of entry” for citizen participation it remains less clear how one cares for nursing relations after projects end. Analysing an ongoing case of a municipality’s efforts to engage citizens in sustainable co-creation towards climate transition, we introduce ”points of exit” as an analytical concept and how the concept of ”aligning actants” is relevant for participatory design, policy development and governance models.

Keywords: participatory design; climate commoning; infrastructural care; Climate governance

1. Introduction

Contemporary local urban governments struggle to approach ambitious environmental goals in the global climate crisis. On the one hand, cities and local government bodies have taken on the role of driving innovation in low carbon transition from a local perspective. This includes technological development, policy work and implementing different sustainability concepts such as circular economies, degrowth, alternative energy sources and many others. These efforts are often explored through project-based initiatives that all together makes for a complex mesh of similar projects in municipalities. On the other hand, an understanding on the necessity to increase just and fair citizen engagement in governmental sustainability initiatives has become fundamental (Ross et al., 2021), the challenge emerges on how processes can become resilient beyond specific project initiatives. Hobson observes how

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sending deliberative participants back into an unsupportive context once deliberation is over can even be more damaging than helpful (Hobson, 2014).

Thus, local government bodies are exploring how they can implement deliberative methods for involving and engaging the public to approach a more informed citizenry and a stronger, more democratic, interaction between municipalities and their citizens. The motivation is two-fold; from a democratic perspective one advocates transparency and accountability in public sector practices and legislation. From a more rational perspective increasing participation from the public paves way for more informed development of public sector services. Also, the increasingly complex challenges for contemporary societies, such as climate change, calls for more collaborative processes and engage citizens in transformative change. For deliberative methods the dialogical format is central, permitting and giving voice to individuals and groups with different backgrounds and values, making it possible to understand and approach more informed decisions, also in everyday life (Abelson et al, 2003; Arendt, 1958). Examples of deliberative features include a broad set of methods such as citizens' juries, planning cells, deliberative polling, consensus conferences, citizens' panels and many others (Abelson et al 2003).

Organisational structures in governments are often thought upon as being inert and that change comes slowly. Mahoney and Thelen's (2009) describes how institutions can be seen as dynamic entities where important change takes place, albeit slowly and incrementally. Huybrechts et al (2017) introduced the concept of *institutioning* for highlighting how institutions are active sites for change and not only an inert backdrop. In municipality's efforts to implement co-creation platforms for sustainable change they interact both with individual citizens but to a large extent also with NGOs being focused on environmental challenges. Sharing the agenda towards climate change between formal governance bodies and citizen groups can be seen as actions of *commoning*. Partelow and Manlosa (2022) defines commoning as the actions by groups with shared interests towards creating shared social and relational processes as the basis of governance strategy.

In this paper we reflect on a case of a municipality's ambitions to innovate the work of co-creating climate change in collaboration with citizens and non-profit interest groups. We describe this as an act of *climate commoning* and how the municipality's ambitions to develop strategies towards more collaborative and interactive formats as a case of *institutioning*. For building long-term relationships beyond specific projects, we here describe their way of working in terms of infrastructuring and aligning actants and how they apply infrastructural care. Especially, we aim to contribute with a discussion on "points of exit", i.e. how participants, be they individual citizens or groups of common interest, must offer opportunities for prolonged participation, learning and acting upon the environmental challenge.

1.1 Co-creating Climate Policy

Citizen engagement in climate change makes for a complex challenge that calls for a multidisciplinary approach. As the challenge of climate change has been problematic to address on a global level more initiatives leading mitigation and adaptation efforts are implemented by lower levels of government such as cities and regions (Rohmsdahl, 2020). Even if the need for engagement from the non-profit/third sector is strong we need a firmer understanding of how such engagement can align with the governance models of cities and regions. To succeed at scale, grass-roots innovation and localism need support from a new environment of policy, finance, infrastructure and technology (Robinson 2014). From that perspective it becomes necessary to develop a configuration of the relationship between governance models and citizenry that is relational rather than transactional (Boyte, 2004) in which climate relevant strategies are co-created in dialogue. Deliberate democracy has also been recognized as a valuable strategy for improving public engagement and decision-making strategies for complex environmental issues (Ney and Verweij, 2015). Consequently, working on a more local scale, everyday practices of urban life and citizen-driven initiatives come more into focus and social cohesion and inclusion becomes parameters in adapting to climate change. Romsdahl observes how experts in the late 90s were calling for more open views of climate change, to expand discussions beyond the dominant idea of emissions reductions; they argued this would allow climate concerns to be incorporated into everyday issues faced by people at local levels (Romsdahl, 2020).

Situated within urgent debates on sustainability transitions, *Klimatomställningar i Samverkan* (lit. *Climate transitions in collaboration*) is a year-long collaboration between Malmö University researchers, local climate activists, and environmental strategists from the Malmö Municipality Environmental Department. The department is one of several specialized municipal agencies that implement national and local policy goals. In Sweden's governance system, these agencies are responsible for delivering services, managing infrastructure, and conducting strategic work—from public health to urban planning—under the political direction of elected city councils.

While the Environmental Department holds a mandate to drive climate transitions across sectors—including collaboration with companies and other public agencies—it faces systemic challenges in engaging citizens. Reaching residents in meaningful ways is resource-intensive and uncertain, with no guarantee that participation will translate into sustained action.

2. Background and Related Work

2.1 Participatory Design and Temporality

Participatory Design (PD) has long been grounded in the orchestration of concrete design events—workshops, prototyping sessions, and encounters that bring together designers, researchers, and those affected by design interventions. Yet, as Saad-Sulonen et al. (2018) argue, such *front-stage* activities represent only fragments of participation's temporal unfolding. PD has traditionally been structured around two temporal orientations: a future-oriented one, where participation serves to co-imagine preferable futures, and a project-based one, where design unfolds within the bounded timeline of a funded initiative (Saad-Sulonen et al., 2018; Simonsen & Robertson, 2012). These dominant framings have shaped how participation is conceptualised—tending to privilege beginnings and moments of intense collaboration while rendering less visible the slower, more ambiguous forms of engagement that precede and succeed such events (Halskov & Hansen, 2015; Light & Akama, 2012).

Recent scholarship has sought to expand this view by attending to participation as processual and extended (Karasti, 2014; Pihkala & Karasti, 2016). Drawing on perspectives from Science and Technology Studies (STS), *infrastructuring* has been introduced to describe design as the continuous alignment and re-alignment of socio-material relations over time (Star & Ruhleder, 1996; Karasti et al., 2010). From this perspective, participation does not merely occur *in* projects but *through* them, as infrastructures of collaboration and care are built, maintained, and transformed.

This temporal expansion destabilises the traditional distinction between *design time* and *use time* (Ehn, 2008; Redström, 2008), repositioning participation as a distributed and evolving practice rather than a discrete event. The introduction of temporal lenses in PD - phasic, emergent, retrospective, prospective, and long-term (Saad-Sulonen et al., 2018) - has provided an analytical vocabulary for engaging with these complexities. The emergent and long-term lenses, in particular, foreground participation as a form of ongoing *becoming* (Stengers, 2011), in which designers and participants are entangled in evolving relations that extend beyond the project's lifespan.

While PD has advanced tools for initiating participation and designing *entry points* for engagement, it has paid less attention to the *exits*—the moments when projects conclude, facilitators withdraw, or institutional attention shifts elsewhere. Attending temporality therefore invites us to reconsider PD not only as a practice of initiating participation but as a politics of sustaining it. Seen through this lens, design becomes a mode of infrastructural care, in which relationships, commitments, and capacities must be continuously maintained, reconfigured, and reimagined over time.

2.2 Governance and Participatory Design for Policy

The usefulness of design methodologies in developing, implementing, and assessing public policy has gained increasing recognition in recent years. This growth is linked to a wider political and academic interest in involving citizens and civil society more directly in policy development and evaluation. *Design for policy* approaches apply design principles and participatory methods to the creation of public policy—emphasising collaborative problem-solving, participatory decision-making, and iterative testing of new governance models (Bason, 2014; Mortati & Schmidt, 2024).

Participatory design, in this context, is not only a method for developing user-centred services but also a means for cultivating democratic capacities within institutions. It provides formats for dialogue and experimentation that enable public sector actors to navigate complexity and ambiguity in policy work (Kimbell & Bailey, 2017). Such approaches allow policy design to become more responsive, situated, and open to learning.

Within the field of climate governance, participatory design has been proposed as a pathway towards *procedural justice*—that is, ensuring that individuals and groups have equal opportunities to influence decision-making processes. Gordon et al. (2022), for example, describe a framework for collaborative climate adaptation based on *Deep Listening*, which provides frontline and Indigenous communities with the means to collect and translate local narratives into institutional languages capable of shaping policy. This framework highlights participation as a reciprocal process of translation between lived experience and bureaucratic structures.

A key strength of participatory design lies in its creative and speculative orientation. By employing visualisation, prototyping, and storytelling methods, participants can explore possibilities that extend beyond established frames of rational argumentation (Mintrom et al., 2024). These approaches make it easier for individuals less familiar with institutional discourse to express situated knowledge, experiences, and values. In doing so, they help redistribute epistemic authority within the design of public policy.

Acknowledging power asymmetries between users, designers, and policymakers remains crucial. As Volkmann et al. (2023) argue, participatory methods can enable marginalised groups to articulate their perspectives and influence decisions—yet this potential depends on how institutional structures accommodate and value such input. Participatory design for policy thus requires careful attention to how participation is staged, sustained, and recognised within organisational contexts. Recent work by Yee, Spencer and Defeyter (2024) is useful here, as it shows how co-design processes situated in local public service contexts can function not only as service improvement, but as a form of designing for policy: that is, as an infrastructuring practice through which local experiments, relationships, and representations can be mobilised to influence policy development beyond the immediate project setting. From this vantage point, design's contribution to governance can be seen not

only in its capacity to produce innovative solutions, but in its ability to mediate relations between diverse actors, knowledges, and temporalities. Design becomes a mode of *institutioning* (Huybrechts et al., 2017)—a practice that intervenes in the ongoing reconfiguration of public institutions. When municipalities engage in co-creation processes, they do not merely invite citizens to participate in predefined structures; they also expose those structures to transformation.

This relational and processual understanding of design situates it within broader discussions of democratic innovation (Smith, 2009) and experimental governance **(**Voß & Bornemann, 2011). In both, the emphasis lies on learning and reflexivity: governance understood not as the control of outcomes, but as the cultivation of shared capacities for adaptation and care.

Together, these perspectives establish the conceptual foundation for the present study. They position *participatory design* as both a methodological and a political resource for municipalities seeking to enact climate transitions in collaboration with citizens and civil society. Within this framework, participation becomes not an end but an infrastructural condition—one that requires ongoing maintenance, negotiation, and renewal.

2.3 Infrastructuring and institutioning

This paper departs from a relational understanding of agency, and of design as a situated and always-already political activity. As has been argued elsewhere (Smedberg, 2022), *infrastructuring* is not simply the building of infrastructures but a form of *care work*: a sustained and situated effort to hold together socio-material constellations that enable certain ways of acting in the world while foreclosing others. In line with this understanding, design is not an isolated intervention, but a practice embedded in historical layers, institutional conditions, and organisational frictions. It is always entangled with the currents of power, legitimacy, and recognition that run through the public sphere. Following this line of thought, the paper does not ask what design can *solve*, but how it can render visible the negotiations and asymmetries already at play—how it can participate in shaping the conditions of collective life without erasing the tensions that inhabit it.

The theoretical framing draws on Science and Technology Studies (STS), Feminist Technoscience, and Participatory Design (PD). These framings converge in their commitment to situated knowledge and to an understanding of the political as something that unfolds not only in moments of decision, but within the ongoing and mundane practices that shape how we live together. The *infrastructuring* practitioner, as conceptualised in earlier work, recognises the limits of their own agency and works instead to build, maintain, and mobilise shared capacities over time. This entails a move from the solitary to the collective, from the product to the relation, and from problem-solving to caring for the conditions of participation.

Building on recent work by Huybrechts, Benesch and Geib (2017), we complement this perspective with the notion of *institutioning*. Institutioning highlights the active and dynamic role of institutions in framing and being framed by participatory processes. It describes how PD operates across micro-, meso- and macro-political levels, shaping and being shaped by the policy frames, institutional action frames, and metacultural frames that govern public life. Rather than treating institutions as inert or monolithic, institutioning invites designers

and researchers to engage them as living infrastructures—porous, contested and open to transformation. In this sense, institutioning expands participation beyond the local and the project-based, recognising the slow, negotiated and multi-directional movements through which design contributes to institutional change.

In a *changing world*—marked by intersecting ecological, technological, and social crises (Huybrechts et al., 2024)—this perspective positions design as both critique and contribution: a practice that navigates between maintaining existing institutional structures and re-articulating them in more caring, just, and sustainable ways. Institutioning, when intertwined with infrastructuring, thus becomes a form of *infrastructural care*: an ongoing negotiation of stability and transformation in the public realm. It foregrounds the relational and temporal dimensions of design work in contexts such as climate governance, where sustaining collective capacity often depends less on discrete interventions than on the long-term reconfiguration of institutional relations.

2.4 The concept of alignments and actants

Cities are most often engaged in a complex ecosystem of projects, initiatives, strategy documents and policies. These relate to citizen engagement and organisational learning in different ways. It seems wise to have a holistic overview of what is already ongoing and to have flexible strategies for entering emerging alliances. Hillgren et al (2013) observe two different management narratives. One, and the most common one, brings forward key concepts such as planning, control, strategy development, common visions and agreements

and heroic managers that make rational choices. The other, that almost seems to be suppressed by the first, can be termed as the complexity narrative that takes a more mundane approach and view innovation as something that can't easily be fixed in models or abstractions and where innovation constantly emerge in everyday life through "ongoing active participation in the exploration of difference and diversity in organisations or communities."

Rather than assessing each project or initiative as monadic entities with clearly identifiable goals it can be a wise strategy to search for overlapping agendas, partially overlapping groups of actors (both among citizens and municipality officers) and potential synergy effects. A strategy of aligning project activities with its stakeholders' already existing interests has in many cases been effective for building collaborative scenarios, including alignment

with projects that the city is already undertaking. While it's common to address matchmaking among diverse sets of human actors, this strategy should involve also non-human actors in the networking ambitions such as projects, technical platforms, institutional instances and not the least policy development and strategies. Including "non-human" actors can be referred to actor-network theory and the French research sociologist Bruno Latour

(1990) for example uses the term actant to describe both human agents and the non-human artifacts.

We can still most often see some clear boundaries between actors, projects, business partners, NGOs and the municipality. But as engagement activities proceeds specific processes, sub-projects and alliances can be formed, and we can think about such a strategy as one of aligning actants. As an "infrastructuring" approach to engagement it can allow a more ongoing and open-ended infrastructure to evolve, both regarding constellations, who will participate, and questions and issues, what to explore and how to do it. It is characterized by a continuous process of building relations with diverse actors and by a flexible allotment of time and resources. This more organic approach to engagement facilitation, here termed aligning actants, the emergence of possibilities along the way and tries to make use of the creative potential in the heterogeneous city, where serendipity could play a role and unexpected and exciting combinations of people could become productive through a continuous matchmaking process.

With these conceptual commitments in place, the paper now turns to the empirical foundations on which it rests. The case study presented here emerged from an ongoing collaboration between researchers, municipal actors and civil society organisations in Malmö and Sweden, within the framework of a policy lab concerned with local climate.

4. Case *Klimatomställningar i Samverkan*

4.1 Malmös strategies and challenges

Malmö has for the last decades had an ambitious climate policy. It has formulated several goals for the year 2030, such as to reduce the emissions from greenhouse gases by 70 % and to reconvert to a 100 % use of renewable energy sources (Malmö Municipality, 2021). In the city's environmental programme, interaction with businesses and civil society as well as citizen participation are presented as key factors for success. To reach these goals, one essential tool is the so called climate contracts, which are social contracts that can be signed by civil society organizations or businesses (Malmö Municipality, 2025A). For a business to sign a contract, it must commit to at least five measures to reduce its climate impact (out of a total of 40). Civil society organizations must commit to one out of four actions. Malmö has the role of facilitating interaction within and between different sectors.

One important reason for involving citizens in climate policy-making is to increase the level of procedural justice. However, the fact that the technical department at Malmö City

often has the mandate in climate policy-making is a challenge as it may lead to a focus on technological issues and a de-prioritization of questions of citizen engagement and social justice. In addition, the climate goals must be reached quickly, which also is a hindrance for broad citizen participation. Moreover, Malmö is a segregated city with large differences between the consumption patterns of the richer and the poorer segments of the population. Some experience energy and transport poverty, while others have high consumption-based emissions, due to private car use and frequent flying. Therefore, it is a challenge to design methods for participation that address the needs of the different segments of the population.

4.2 Case study Overshoot festival

The project responds to this challenge by creating a shared space where municipal officials and grassroots actors can collaborate, exchange knowledge, and experiment with new forms of climate engagement. Instead of emphasizing individual behaviour change, the initiative foregrounds collective capacities and networks within civil society organizations. These organizations span a wide range—from climate justice groups and neighbourhood associations to cultural collectives and sports clubs—and represent a crucial but often underutilized infrastructure for civic engagement in Sweden.







FIG 1-3: PHOTOS FROM OVERSHOOT FESTIVAL, MALMÖ 2025. (PHOTOS: HÅLLPUNKT)

The design-driven process has taken form through the co-organization of Overshoot Festival, a citizen-led event that functions simultaneously as a public intervention and a prototype for climate-related collaboration. Beyond the festival, the project has included workshops and dialogical encounters with over fifty organizations, weaving together the Environmental Department's institutional practices with activist forms of engagement.

Although formally limited to one year, the collaboration builds on three earlier research projects with a similar constellation of actors. This continuity signals an ongoing process of *infrastructuring*—a long-term, open-ended attempt to reconfigure relations between state and civil society around shared commitments to environmental justice.

4.3 Anecdotes as empirical windows

In this paper, we present three anecdotes from our collaboration between activists and public sector workers in Malmö. Together, they open up the texture of collaboration between municipal officials and citizens engaged in climate transition work. Rather than treating these moments as incidental, we approach anecdotes as condensed narratives that crystallize broader dynamics of power, affect, and recognition (Smedberg, 2022). Anecdotes do not claim to represent the whole; rather, they function as empirical windows—*puncta* that interrupt smooth accounts and reveal the frictions, negotiations, and alignments that constitute infrastructuring work.

Infrastructuring is never only about strategic decisions or formal plans; it unfolds in micro-moments where heterogeneous elements—people, roles, organizations, and material artefacts—are brought into relation. Anecdotes help us trace these situated alignments,

showing how they are partial, fragile, and affectively charged.

The first anecdote stems from the very first encounter between Participant 1, whom we'll call Jenny, an environmental strategist from the municipality, and Participant 2, whom we'll call Alex an activist affiliated with the association *Hållpunkt.* Alex recalled entering the municipal meeting room acutely aware that their activist background might mark them as an outsider. The feeling was sharpened by a laptop sticker reading *Genocide Song Contest* —a critique of the Eurovision Song Contest then hosted in Malmö. Facing Jenny and her municipal colleagues, Alex reflected:

> *"It is like I brought a conflict into the room that I did not intend to put on the table. These small markers make it tangible that we sit together in situations that carry a lot of tension."*

Jenny, in turn, reflected on the representational burden of being seen as *the municipality*:

> *"One of my concerns was that people might expect me to carry all of Malmö on my shoulders. But decisions are made far above my mandate. In fact, I was also critical of Eurovision at the time."*

What appears as a minor detail—a sticker on a laptop—becomes a lens into the complexities of collaboration. It highlights how actors carry symbolic affiliations and institutional expectations into the room, and how these can both obstruct and enable dialogue. Aligning actants here involves the subtle navigation of conflicting affective registers: irony, critique, and institutional responsibility.

The second anecdote takes us out of the municipality's meeting rooms and into the activist terrain of the *Overshoot Festival*. There, Jenny and a colleague from the environmental department set up a small stand, serving coffee and inviting visitors to discuss Malmö's climate initiatives. Their presence was meant as a gesture of openness—to meet citizens on their own ground, outside formal hearings or consultation processes.

Yet what unfolded revealed the difficulty of stepping into such spaces as a public servant. Among the visitors was a woman who returned repeatedly, pressing the strategists with pointed questions about cement production in Malmö. The topic, while relevant, lay outside their specific mandate. Her persistence interrupted other conversations, creating visible discomfort. From the strategists' perspective, the encounter was stressful: they lacked both the information and the authority to respond. Attempts to redirect the discussion only

intensified her frustration, producing a misalignment of expectations—between a citizen seeking accountability and officials constrained by institutional boundaries.

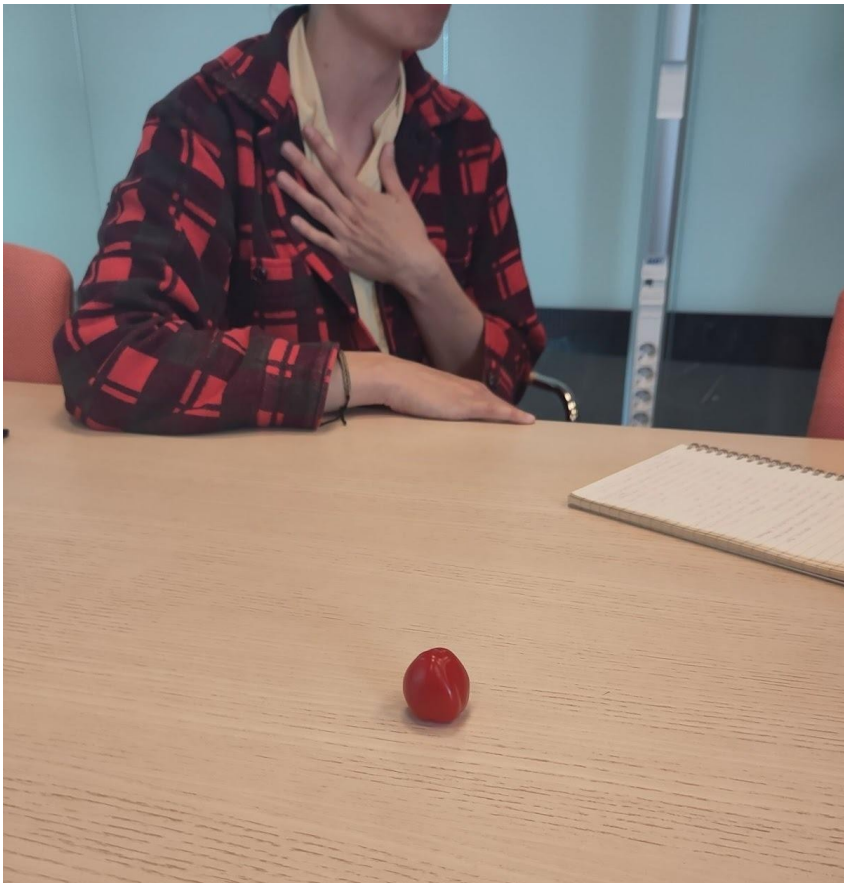
Later, back at the office, Jenny's colleague questioned whether they should have been there at all:

> *"The municipality offered coffee and cake—shouldn't you have been networking instead? Wouldn't it be better getting to know the organizations* [exhibiting at the festival] *rather than saying 'hi' to a thousand Malmö residents?"*

The comment captures a tension internal to the municipality itself: whether participation means broadcasting messages to the public or building relationships with intermediaries who already mobilize civic engagement. This internal debate, like the festival encounter, illuminates the unstable alignments that public officials must continually negotiate when stepping outside institutional routines.

The third anecdote centres on Joakim, another climate strategist from Malmö municipality, who participated in a *study circle.* Study circles are often used in popular education and civic organizations in Sweden, a central part of *Folkbilding*. Unlike a formal workshop, the study circle is self-organizing and conversational, emphasizing mutual learning rather than instruction. In this circle, organized within our project, Joakim participated as one among others rather than as an institutional representative. He shared the communication dilemmas the city faces when addressing climate transition: they are not allowed to sound too alarmist, for fear of scaring the public, yet cannot appear indifferent. "The responsibility to reach everyone," he reflected, "often means we end up reaching almost no one."

Among the participants was a climate psychologist who referenced a study (Andre et al., 2024) showing that while most people are willing to make sacrifices for the climate, they believe that others are not—creating a false sense of isolation. The discussion became personal: one woman recounted her practice of "*dumpstring"* (collecting discarded food from grocery bins) and followed the life story of a single tomato—grown in Spain, transported across Europe, discarded after its expiry date in Sweden, and finally rescued from a dumpster. For Joakim, hearing such stories was reassuring: the municipality's struggle to motivate change was mirrored by citizens' own feelings of being alone in their efforts.



PARTICIPANT WITH TOMATO, JUNE 2025 (PHOTO: ALICIA SMEDBERG)

These three anecdotes—each from different sites of collaboration—foreground how infrastructuring is animated by moments of alignment and misalignment among human and institutional actants. They expose the affective, symbolic, and organisational negotiations through which participation unfolds, and set the stage for our subsequent discussion of *aligning actants* as a central analytical theme.

5. Discussion Conceptual Framing: Aligning Actants och climate commoning

Across the three anecdotes, collaboration between activists and municipal officials unfolds in small, uncertain gestures rather than in moments of delivery. These encounters—whether in the municipal meeting room, the festival park, or the study circle—reveal what it means for public servants to *align actants*: to bring together people, roles, materials, and affects in ways that allow collaboration to continue, even when consensus cannot be reached.

In the first anecdote, a sticker on a laptop quietly disrupts the neutrality of the municipal meeting room. Alex's awareness of being "the activist" and Jenny's worry about representing "the whole city" expose how affective and symbolic alignments must be constantly negotiated. The moment is fragile but generative: it opens the possibility for mutual recognition across institutional boundaries. This is the work of *aligning actants*—not the achievement of harmony, but the maintenance of relation in difference.

At the Overshoot Festival, this practice of alignment becomes more public and more difficult. The tension between Elin's intended openness and the citizen's insistent critique illustrates the precariousness of municipal participation in civic spaces. Later, her colleague's comment —questioning whether "serving coffee to a thousand Malmö residents" was worthwhile— captures an internal conflict within the municipality itself. As Boztepe et al. (2023) note, public servants who engage with design and participation must constantly navigate *tensions of positioning*: between being insiders and outsiders, between representing the municipality and experimenting with it. In this light, Jenny's gesture of service at the festival is not inefficiency but infrastructural care. By choosing to stay in the encounter rather than manage it away, she sustains a connection between the city and its citizens, however partial.

The third anecdote, centering on Joakim in the study circle, extends this relational work into a space of shared reflection. By joining as one participant among others, Joakim suspends the asymmetry that usually structures municipal communication. His willingness to speak about institutional constraints—how the city cannot be "too alarmist" nor "too calm" when talking about climate—invites others into the complexity of public responsibility. The study circle, as a format, facilitates what Light and Seravalli (2019) describe as *co-learning*: the process through which municipalities can rediscover their role as *caring platforms*. Within the circle, care circulates horizontally rather than hierarchically. The climate psychologist's story about collective misperception—our sense of being "alone" in caring—illustrates the psychological terrain of climate governance. When Joakim hears the story of the "dumpstered tomato," he becomes part of a shared narrative that reconnects personal and institutional efforts. This, too, is alignment: a municipal actor finding solidarity in the affective commons of climate concern.

Seen together, these moments trace a municipal practice of *infrastructuring through care*. They show how civil servants, activists, and citizens engage in the ongoing work of keeping relations open across organisational, social, and affective boundaries. Following Light and Seravalli (2019), we might see this as an attempt to repair the municipality's capacity to act as a *caring infrastructure* in a time when public institutions are pressured to perform efficiency rather than maintain relations. The officials' work is not to lead or deliver, but to hold space—to stay in contact with the discomforts that accompany collaboration.

When read through Perkins' (2021) notion of *climate commoning*, these alignments become visible as forms of collective care for the conditions of collaboration itself. Climate commoning is not limited to managing shared resources; it also includes sustaining the social and institutional relationships that make collective action possible. In this sense, aligning

actants becomes a mode of commoning: an everyday practice through which public servants, activists, and citizens co-produce the infrastructures of trust and belonging that climate transitions require.

Rather than large-scale transformation, what emerges is a quieter politics of maintenance. The municipality's role is not to orchestrate participation from above, but to participate *in* the commons it helps sustain—to keep aligning, listening, and caring, even when outcomes are uncertain. Such moments of alignment, however fleeting, reveal how climate governance can be enacted through situated acts of infrastructural care: through the municipality's willingness to be part of, rather than apart from, the shared work of commoning.

The framework of infrastructural care leads to another conception of justice than what has been common in the debate within ethics and social science. Feminists often regard care as a primary concept, although not necessarily in opposition to justice (Held, 2005, 2015). Care ethicists who endorse the use of the justice concept want to reconceptualize it as a relational and context-sensitive practice, rather than an application of abstract and universal principles (Lanphier, 2021). They advocate a special concern for the needs and lived experiences of marginalized and dependent groups.

From the perspective of care ethics, municipalities and other more privileged groups have a responsibility to include marginalized and disadvantaged groups in collaborative processes and in decision-making, especially as climate policies may exacerbate already existing inequalities (Markkanen and Anger-Kraavi, 2019). At the same time, climate policy-making at the municipal level needs to address the high consumption-based emissions of the richer segments of the population. As mentioned, in a segregated city like Malmö it is challenging to design methods for participation that are relevant for different segments of the population.

5.1 Infrastructural care as exit points

If we see ”points of entry” as the local governments readiness for inviting citizens to dialogue and participation in the efforts towards climate transition, we can observe that the municipality has a variety of tools and methods as well as experience of implementing this interaction in different formats. A strategic vision and roadmap for co-creation towards climate change is also in place, internally developed in the municipality. It remains less clear what ”points of exit” that are offered once different initiatives end. The concept of infrastructuring is helpful for more long-term resilient co-creation networks but the actual mechanisms for sustaining have received less attention than the aspects of ongoing participation through different deliberate methods.

We propose that focusing on these points of exit as an analytical lens would be fruitful further research in participatory design, and especially in the context of government-citizen interaction. Such a focus can be understood as an instance of infrastructural care. In the case of Malmö we can see seeds of how this care can be performed in practice in the list below.

1. *The aspect of aligning actants*, i.e. overlooking the entangled landscape of different projects, initiatives, steering documents, NGOs and individuals provides a position in which the municipality can act as matchmaker of similar communities of practice. We can observe how the environmental officers constantly search for synchronizing opportunities, thus offering participants continued participation/engagement through entering into new forums for participation that can build upon previous experience. This can be seen as a careful interweaving of values, positions and practices that can go beyond short-term temporalities, thus facilitating mobilization of new network constellations.

2. *Balancing leadership with facilitating and stepping back from leadership*. There is a need for a strong internal common vision in the municipality, which is provided by strategic visions. However, we see a need for also having sensibility for letting the vision to at times turn into a quieter politics of maintenance. For this to happen, the government body must also at times let go of the leading role, handing it over to other actors like in the case of supporting the Overshoot festival instead of arranging themselves. In the anecdote of Joakim taking part as just another participant in the studying circle while at the same time contributing with his expertise we see another example of balancing leadership with suspending the asymmetry of power relations.

3. *Moving from top-down defined goal targeting to mundane contexts*. The movement of ”stepping back from leadership” is related to another movement; that of shifting focus from the general eagle-eye perspective of the strategic documents goal to decrease emissions to translations into sustainable everyday life-styles. There is a need to understand how to combine technical support and environmental sustainability efforts with social sustainability. A shift also in semantics, moving from the dominance of general fact-based language to also include anecdotes as empirical windows is another way of supporting this direction.

4. *Identifying and supporting collective power and providing learning opportunities*. The Overshoot festival illustrates the power of collaborative ”gathering of the tribes” toward environmental change. Whereas facilitating the festival can be seen as an act of aligning actants, the presence of the environmental officers with expertise in climate issues makes it possible for learning opportunities. Further, co-learning on everyday life-styles occurred also in the study circle and support the mentioned climate psychologist’s argument that it’s needed to oppose the ”

false sense of isolation", i.e. that one does not recognize that huge numbers of people are engaged through everyday actions and that inter-individual learning is possible. From this perspective the concept of energy commons becomes valuable.

5. *Continuous feedback loops as a way of maintaining engagement.* It is reasonable to provide feedback to participating citizens and NGOs on how their participation affects (or not) the governing bodies decision-making. Public policy labs could be one way of mitigating low effects of participation. Providing offerings for prolonged participation in upcoming events and activities and feedback on results of the city's effort towards climate change is a way of not only "exiting" but "continuing".

6. Conclusion

This paper has explored how municipalities can engage in *climate commoning*—a collective, care-oriented approach to governing environmental transitions. Through the case of Malmö, we have shown how collaboration between public officials, researchers, and civil society unfolds not primarily through grand strategies or innovations, but through the slow and situated work of maintaining relations across institutional and civic boundaries.

By bringing together the concepts of *infrastructuring* and *institutioning*, we have described design as a practice of *infrastructural care*: an ongoing negotiation between stability and transformation within the public realm. This perspective invites attention to how design can both sustain and reconfigure the institutional conditions that make participation possible, particularly in contexts marked by complexity, uncertainty, and uneven distributions of power.

Our discussion of *exit points* extends this argument by emphasising that the durability of participatory processes depends not only on how they begin, but on how they end. Designing for exit—ensuring that knowledge, relationships, and capacities can continue to circulate beyond the lifespan of a project—is essential to building resilient and just forms of public engagement. In this sense, climate commoning becomes a practice of staying in relation: keeping the infrastructures of care open, even as specific collaborations conclude.

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